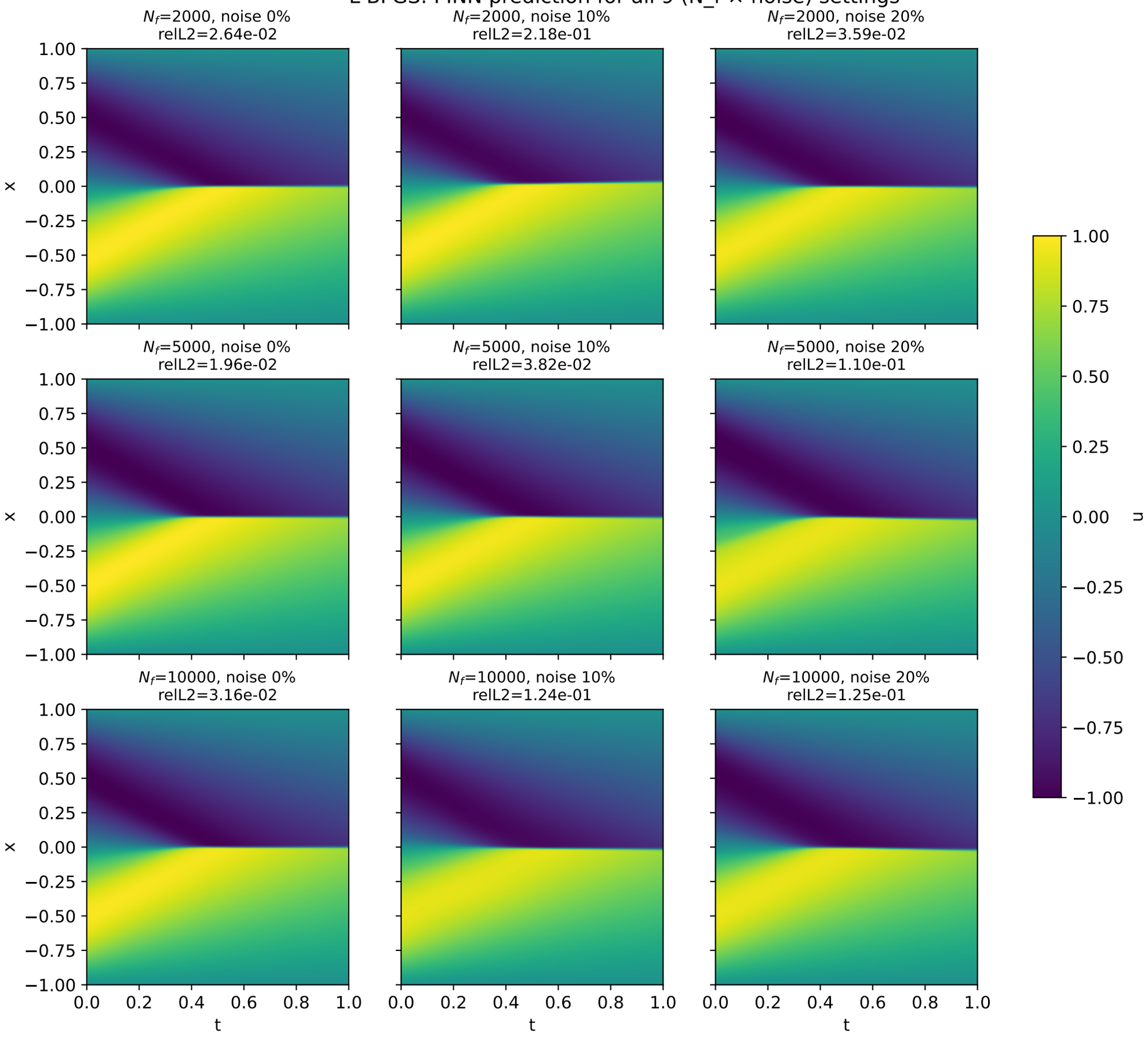

L-BFGS: PINN prediction for all 9 (N_f × noise) settings
N_f=2000, noise 0%
relL2=2.64e-02
N_f=2000, noise 10%
relL2=2.18e-01
N_f=2000, noise 20%
relL2=3.59e-02
N_f=5000, noise 0%
relL2=1.96e-02
N_f=5000, noise 10%
relL2=3.82e-02
N_f=5000, noise 20%
relL2=1.10e-01
N_f=10000, noise 0%
relL2=3.16e-02
N_f=10000, noise 10%
relL2=1.24e-01
N_f=10000, noise 20%
relL2=1.25e-01
x
t
u
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.0
0.2
0.4
0.6
0.8
1.0